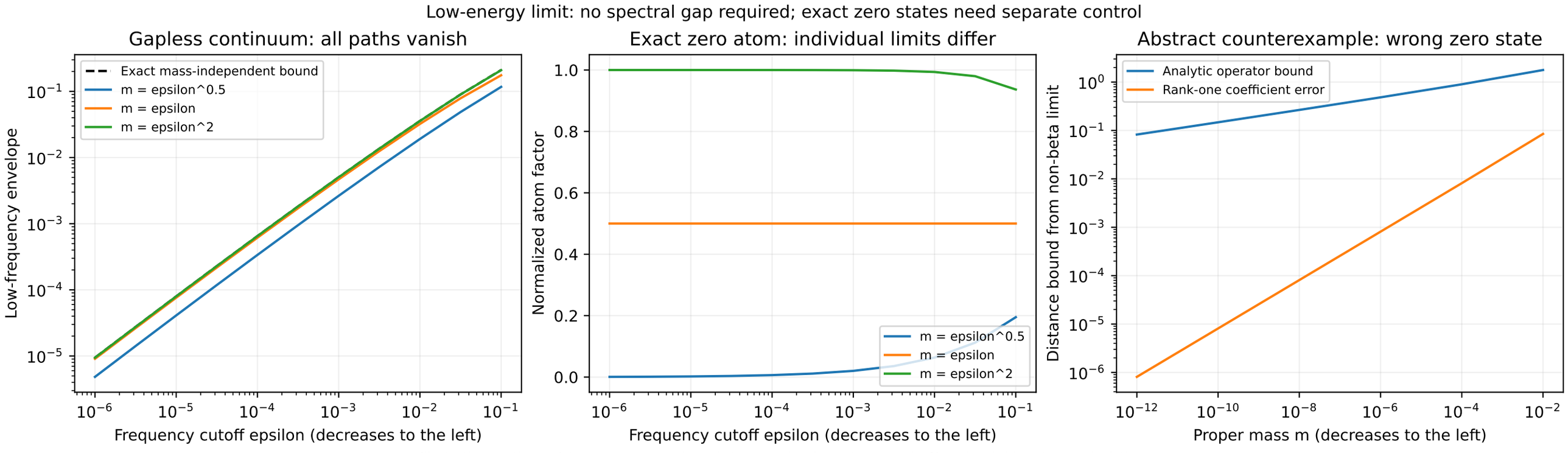
Low-energy limit: no spectral gap required; exact zero states need separate control
Gapless continuum: all paths vanish
Exact mass-independent bound
m = epsilon^0.5
m = epsilon
m = epsilon^2
Low-frequency envelope
Frequency cutoff epsilon (decreases to the left)
Exact zero atom: individual limits differ
Normalized atom factor
m = epsilon^0.5
m = epsilon
m = epsilon^2
Frequency cutoff epsilon (decreases to the left)
Abstract counterexample: wrong zero state
Analytic operator bound
Rank-one coefficient error
Distance bound from non-beta limit
Proper mass m (decreases to the left)
Illustrative continuum/atom measures and an abstract nonlocal counterexample. None is a bridge stress calculation.